

Seminar handout

# O-Minimal Structures Seminar

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# 1 Cylindrical algebraic decomposition

**Theorem 1.1 Thom lemma.** Let  $f_1, \dots, f_k \in \mathbb{R}[T]$  be nonzero polynomials such that if  $f'_i \neq 0$ , then  $f'_i \in \{f_1, \dots, f_k\}$ . Let  $\epsilon : \{1, \dots, k\} \rightarrow \{-1, 0, 1\}$ , and put

$$A_\epsilon := \{t \in \mathbb{R} : \text{sign}(f_i(t)) = \epsilon(i), i = 1, \dots, k\}, \text{ a subset of } \mathbb{R}.$$

Then  $A_\epsilon$  is empty, a point, or an interval.

Moreover, If  $A_\epsilon \neq \emptyset$ , then its closure is given by

$$\text{cl}(A_\epsilon) = \{t \in \mathbb{R} : \text{sign}(f_i(t)) \in \{\epsilon(i), 0\}, i = 1, \dots, k\}.$$

If  $A_\epsilon = \emptyset$ , then  $\text{cl}(A_\epsilon)$  is empty or points.

**Proof.** By induction on  $s$ . The Lemma holds trivially for  $s = 0$ . Let  $f_1, \dots, f_s, f_{s+1} \in R[\mathbf{x}] \setminus \{0\}$  be polynomials such that if  $f'_k \neq 0$ , then  $f'_k \in \{f_1, \dots, f_{s+1}\}$ . Without loss of generality we assume that  $\deg(f_{s+1}) = \max\{\deg(f_k) : 1 \leq k \leq s+1\}$ .

Let  $\epsilon' : \{1, \dots, s, s+1\} \rightarrow \{-1, 0, 1\}$  and  $\epsilon : \{1, \dots, s\} \rightarrow \{-1, 0, 1\}$  the restriction.

Note that

$$A_{\epsilon'} = A_\epsilon \cap \{x \in R : \text{sign}(f_{s+1}(x)) = \epsilon'(s+1)\}.$$

By induction  $A_\epsilon$  is empty, a point, or an interval.

If  $A_\epsilon$  is empty or a point, then obviously so is  $A_{\epsilon'}$  and the other property follows immediately by induction hypothesis on  $A_\epsilon$ .

Assume  $A_\epsilon$  is an interval. Now  $f'_{s+1} = 0$  or  $f'_{s+1} \in \{f_1, \dots, f_s\}$ . So by definition of  $A_\epsilon$ ,  $f'_{s+1}$  has constant sign on  $A_\epsilon$ . Therefore  $f_{s+1}$  is either strictly increasing, or strictly decreasing or constant on  $A_\epsilon$ .

Consider  $A_\epsilon = (a, b)$ . There are three cases depending on  $\epsilon'(s+1)$ :

**Case 1.**  $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) > 0\}$ .

**Case 2.**  $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) < 0\}$ .

**Case 3.**  $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) = 0\}$ .

If  $A_{\epsilon'} = \emptyset$  there is nothing to prove.

Assume  $A_{\epsilon'} \neq \emptyset$ . If  $f_{s+1}$  is constant on  $A_\epsilon$  then  $f_{s+1}$  is a constant polynomial  $f_{s+1}(x) = c \neq 0$ . So  $A_{\epsilon'}$  is empty or  $A_{\epsilon'} = (a, b)$  depending on whether  $\text{sign}(c) = \epsilon'(s+1)$ .

Assume now  $f_{s+1}$  strictly increasing on  $A_\epsilon$  and  $A_{\epsilon'} = \{x \in (a, b) : f_{s+1}(x) > 0\} \neq \emptyset$ . Let  $x_0 = \inf\{x \in (a, b) : f_{s+1}(x) > 0\}$ . Since  $f_{s+1}$  is strictly increasing it follows that  $f_{s+1}(x) > 0 \forall x \in (a, b)$  with  $x > x_0$ . So  $A_{\epsilon'} = (x_0, b)$  and its closure is  $[x_0, b] = A_{\epsilon'}$ . The other cases are treated similarly. □

**Notation.**  $X$  is a nonempty topological space,  $E$  a ring of continuous real-valued functions  $f : X \rightarrow \mathbb{R}$ , the ring operations being pointwise addition and multiplication, with multiplicative identity the function on  $X$  that takes the constant value 1.

For  $f(T)$  as above and  $z \in X$ , we write  $f(z, T)$  for the polynomial

$$f(z, T) = f_0(z) + f_1(z)T + \dots + f_d(z)T^d \in \mathbb{R}[T].$$

Let us say that a polynomial  $f(T) \in E[T]$  has *zero-free leading coefficient* if

$$f(T) = c_d T^d + c_{d-1} T^{d-1} + \cdots + c_0$$

with  $c_i \in E$ , such that  $c_d$  has no zero on  $X$ .

**Definition 1.1.** Call a set  $A \subseteq X$  an  $E$ -set if  $A$  is a finite union of sets of the form

$$\{x \in X : f(x) = 0, g_1(x) > 0, \dots, g_k(x) > 0\}, \text{ with } f, g_1, \dots, g_k \in E.$$

Note that the  $E$ -sets form a boolean algebra of subsets of  $X$ . unions, intersections and taking complements.

**Definition 1.2.** If  $X = \mathbb{R}^m$ , and  $E = \mathbb{R}[x_1, \dots, x_m]$  is the ring of real polynomial functions on  $X$ , then the  $E$ -sets are exactly the semialgebraic subsets of  $\mathbb{R}^m$ .

**Lemma 1.1 Lemma 2.** Let  $G(A, T) = A_0 + A_1 T + \cdots + A_d T^d \in \mathbb{Z}[A, T]$ ,  $A = (A_0, \dots, A_d)$ , be a general polynomial of degree  $d$ , and  $e \in \{0, \dots, d, \infty\}$ . Then the set

$$\{a \in \mathbb{R}^{d+1} : G(a, T) \text{ has exactly } e \text{ distinct complex zeros}\}$$

is a finite union of sets of the form

$$\{a \in \mathbb{C}^{d+1} : p_1(a) = \cdots = p_k(a) = 0, q(a) \neq 0\},$$

where  $p_i, q \in \mathbb{Z}[A]$ .

**Proof.** The cases  $d = 0$  or  $e \in \{0, \infty\}$  are trivial. Let  $d > 0, e \in \{1, \dots, d\}$ , and  $a = (a_0, \dots, a_d) \in \mathbb{C}^{d+1}, a_d \neq 0$ .

Let  $g = \text{degree of GCD}(G(a, T), \frac{\partial G}{\partial T}(a, T))$  in  $\mathbb{C}[T]$ .

Then the number of distinct complex zeros of  $G(a, T)$  is  $d - g$ , and the degree of  $\text{LCM}(G(a, T), \frac{\partial G}{\partial T}(a, T))$  is  $2d - g - 1$ .

Hence the condition is that  $G(a, T)$  has at most  $e$  distinct zeros, which is equivalent to  $d - g \leq e$ , that is, to  $2d - g - 1 \leq d + e - 1$ .

The last condition is equivalent to the condition: (\*) There exist  $q(x, T) = x_0 + x_1 T + \cdots + x_{e-1} T^{e-1}$  and

$$r(x, T) = x_e + x_{e+1} T + \cdots + x_{2e} T^e,$$

with

$$x = (x_0, \dots, x_{2e}) \in \mathbb{C}^{2e+1} \setminus 0,$$

such that

$$G(a, T)q(x, T) = \frac{\partial G}{\partial T}(a, T)r(x, T).$$

This equality can be rewritten as

$$G(a, T)q(x, T) - \frac{\partial G}{\partial T}(a, T)r(x, T) = \beta_0(a, x) + \beta_1(a, x)T + \cdots + \beta_{d+e-1}(a, x)T^{d+e-1}$$

where  $\beta = (\beta_0, \dots, \beta_{d+e-1}) : \mathbb{C}^{d+1} \times \mathbb{C}^{2e+1} \rightarrow \mathbb{C}^{d+e-1}$  is a bilinear function.

So (\*) is equivalent to the condition  $\beta_0(a, x) = \cdots = \beta_{d+e-1}(a, x) = 0$  has nonzero solution  $x \in \mathbb{C}^{2e+1}$ . The last condition is equivalent to the vanishing of all  $(2e+1)$ -minor of the matrix of the linear map  $\beta(a, \cdot)$ . Note that each of the minors is a polynomial in  $a_0, \dots, a_d$  with coefficients in  $\mathbb{Z}$ . Therefore, for each  $d' \leq d$ , the set  $M_e^{d'} = \{a \in \mathbb{R}^{d+1} : G(a, T) \text{ is of degree } d' \text{ and has at most } e \text{ distinct complex zeros}\}$  is the intersection of the set  $\{a \in \mathbb{R}^{d+1} : a_d = \cdots = a_{d'+1} = 0, a_{d'} \neq 0\}$  with the zero set of certain polynomials in  $\mathbb{Z}[A]$ . So  $\{a = (a_0, \dots, a_d) \in \mathbb{R}^{d+1} : G(a, T) \text{ has exactly } e \text{ complex zeros}\} = \bigcup_{d'=0}^d M_e^{d'} \setminus M_{e-1}^{d'}$  a semi-algebraic set.  $\square$

**Corollar 1.1.** As a consequence, for every  $f \in \mathbb{R}[X_1, \dots, X_n, T] = \mathbb{R}[X][T]$ ,

$$f(X_1, \dots, X_n, T) = a_0(X) + a_1(X)T + \cdots + a_d(X)T^d,$$

the set

$$\{x \in \mathbb{R}^n : f(x, T) \text{ has exactly } e \text{ distinct complex zeros}\}$$

is a semi-algebraic subset of  $\mathbb{R}^n$ .

**Proof.** Since  $f(x, T) = G(a_0(x), \dots, a_d(x), T)$ , the corollary follows from Lemma 2.  $\square$

**Theorem 1.2 Cylindrical algebraic decomposition.** Let  $f_1(T), \dots, f_M(T) \in E[T]$ . Then the list  $f_1, \dots, f_M$  can be augmented to a list  $f_1, \dots, f_N$  in  $E[T]$  ( $M \leq N$ ), and  $X$  can be partitioned into finitely many  $E$ -sets  $X_i$  ( $1 \leq i \leq k$ ) such that for each connected component  $C$  of each  $X_i$  there are continuous real-valued functions  $\xi_{C,1} < \cdots < \xi_{C,m(C)}$  on  $C$  with the following two properties:

- i) Each function  $f_n$  ( $1 \leq n \leq N$ ) has constant sign ( $-1, 0$ , or  $1$ ) on each of the graphs  $\Gamma(\xi_{C,j})$  ( $1 \leq j \leq m(C)$ ) and on each of the sets  $(\xi_{C,j}, \xi_{C,j+1})$  ( $0 \leq j \leq m(C)$ ), where  $\xi_{C,0} := -\infty$  and  $\xi_{C,m(C)+1} := +\infty$  are constant functions on  $C$  by convention;
- ii) Each of the sets  $\Gamma(\xi_{C,j})$  and  $(\xi_{C,j}, \xi_{C,j+1})$  from (1) is of the form

$$\{(x, t) \in C \times \mathbb{R} : \text{sign}(f_n(x, t)) = \epsilon(n) \text{ for } n = 1, \dots, N\}$$

for a suitable sign condition  $\epsilon : \{1, \dots, N\} \rightarrow \{-1, 0, 1\}$ .

**Proof.** Let  $d = \max\{\deg_T(f_i), i = 1, \dots, p\}$ . Augment  $f_1, \dots, f_p$  to

$$\{f_1, \dots, f_{p+q}\} = \left\{ \frac{\partial^\nu f_i}{\partial T^\nu} : 1 \leq i \leq p, 0 \leq \nu \leq d \right\}.$$

For each  $\Delta \subset \{1, \dots, p\} \times \{0, \dots, d\}$ , and  $e \in \{0, \dots, pd^2\} \cup \{\infty\}$ , put

$$f_\Delta(T) = \prod_{(i,\nu) \in \Delta} \frac{\partial^\nu f_i}{\partial T^\nu} \in \mathbb{R}[X][T], \text{ and}$$

$$A_{\Delta,e} = \{x \in \mathbb{R}^n : f_\Delta(x, T) \text{ has exactly } e \text{ complex zeros}\}.$$

By Lemma 2,  $A_{\Delta,e}$  is a semi-algebraic set. For a given  $\Delta$  the family  $\{A_{\Delta,e} : e \text{ varies}\}$  forms a partition of  $\mathbb{R}^n$ . Since the class of semi-algebraic sets is a boolean algebra we can

find a partition (the intersection of the partitions)  $\mathbb{R}^n = S_1 \cup \cdots \cup S_k$ , where each  $S_i$  is semi-algebraic such that each set  $A_{\Delta,e}$  is a union of the  $S'_i$ s. We will prove that  $f_1, \dots, f_{p+q}$  and  $S_1, \dots, S_k$  satisfy the conclusion of the theorem.

For each connected component  $C$  of  $S_i$  put

$$\Delta(C) = \left\{ (i, \nu) : \frac{\partial^\nu f_i}{\partial T^\nu} \neq 0 \text{ on } C \times \mathbb{R} \right\}.$$

By Lemma 3, there exist continuous functions  $\xi_{C,1} < \cdots < \xi_{C,r}(C)$  on  $C$  such that  $\{(x, t) \in C \times \mathbb{R} : f_\Delta(C) = 0\} = \Gamma(\xi_{C,1}) \cup \cdots \cup \Gamma(\xi_{C,r}(C))$ .

**Check (i):** If  $(i, \nu) \notin \Delta(C)$  then  $\frac{\partial^\nu f_i}{\partial T^\nu} \equiv 0$  on the sets given in (i).

If  $(i, \nu) \in \Delta(C)$ , then  $C \subset A_{\{(i,\nu)\},e}$ , for certain  $e \in \{0, \dots, d\} \cup \{\infty\}$  and the number of real zeros of  $\frac{\partial^\nu f_i}{\partial T^\nu}(x, T)$  is independent of  $x \in C$ .

Since  $\frac{\partial^\nu f_i}{\partial T^\nu}$  is a factor of  $f_\Delta(C)$ , by Lemma 3, the zeros of  $\frac{\partial^\nu f_i}{\partial T^\nu}(x, T)$ , for  $x \in C$ , must be among the  $\xi_{C,j}(x)$ 's. Since  $C$  is connected, (i) is checked.

**Check (ii):** Let  $B$  be one of the sets in (i). By (i),  $\epsilon(i, \nu) = \text{sign}(\frac{\partial^\nu f_i}{\partial T^\nu}|_B)$  is well-defined. Put

$$B' = \left\{ (x, t) \in C \times \mathbb{R} : \text{sign} \left( \frac{\partial^\nu f_i}{\partial T^\nu}(x, t) \right) = \epsilon(i, \nu), 1 \leq i \leq p, 0 \leq \nu \leq d \right\}.$$

Clearly  $B \subset B'$ . If  $B \neq B'$  then exist  $(x, t') \in B' \setminus B$ ,  $(x, t) \in B$  (say  $t < t'$ ). Thom's lemma implies that  $\{r \in \mathbb{R} : (x, r) \in B'\}$  is connected, so  $\{x\} \times [t, t'] \subset B'$ . Since  $(x, t) \in B$ ,  $(x, t') \notin B$ ,  $f_\Delta(c)$  must change sign on  $\{x\} \times [t, t']$ . But  $f_\Delta(c)$  is a product of  $\frac{\partial^\nu f_i}{\partial T^\nu}$ , so  $f_\Delta(c)$  cannot change sign on  $B'$ , contradiction. Therefore  $B = B'$ . □

**Theorem 1.3 Lojasiewicz.** The number of connected components of a semi-algebraic set is finite, and each of the components is also semi-algebraic.

**Corollar 1.2.** If  $(X, E)$  has the Lojasiewicz property, then  $(X \times \mathbb{R}, E(T))$  also has the Lojasiewicz property, and the image of any  $E[T]$ -set  $S \subseteq X \times \mathbb{R}$  under the projection map  $X \times \mathbb{R} \rightarrow X$  is an  $E$ -set.

**Proof.** Write an  $E[T]$ -set  $S$  as a finite union of sets of the form

$$\{(x, t) \in X \times \mathbb{R} : f(x, t) = 0, g_1(x, t) > 0, \dots, g_k(x, t) > 0\}$$

with  $f, g_1, \dots, g_k \in E[T]$ . Let  $f_1, \dots, f_M$  be a list of all the polynomials in  $E[T]$  that are involved in such a definition of  $S$ . Now apply the theorem to this list. □

**Theorem 1.4 Tarski-Seidenberg Theorem.** The  $\mathbb{R}[T_1, \dots, T_m]$ -sets in the space  $\mathbb{R}^m$  are exactly the semialgebraic subsets of  $\mathbb{R}^m$ . For  $m = 0$  the space  $\mathbb{R}^m$  has only one point, so the pair  $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$  obviously has the Lojasiewicz property for  $m = 0$ ; hence by induction on  $m$ , using (2.9), it follows that  $(\mathbb{R}^m, \mathbb{R}[T_1, \dots, T_m])$  has the Lojasiewicz property for each  $m$ , and that the image of a semialgebraic subset of  $\mathbb{R}^{m+1}$  under the projection map  $\mathbb{R}^{m+1} \rightarrow \mathbb{R}^m$  is a semialgebraic subset of  $\mathbb{R}^m$ .

We can now draw the following conclusion about the model-theoretic structure  $(\mathbb{R}, <, 0, 1, +, -, \cdot)$ , which we will call “the ordered field of real numbers”.

**Corollar 1.3.** The sets  $S \subseteq \mathbb{R}^m$  ( $m = 0, 1, 2, \dots$ ) that are definable using constants in the ordered field of real numbers are exactly the semialgebraic sets. The ordered field of real numbers is o-minimal.

**Proof.** semialgebraic sets form a structure on  $(\mathbb{R}, <)$ . Clearly the primitives of the ordered field of reals are semialgebraic, the semialgebraic sets are definable using constants in the ordered field of real numbers. Finally, the semialgebraic subsets of  $\mathbb{R}$  are clearly the finite unions of intervals and points. The desired result follows.  $\square$

## 2 noether normalization

### 2.1 noether normalization

Let  $f(X_1, \dots, X_n)$  be a nonzero polynomial of degree  $d$  over a field  $K$  of characteristic 0, say

$$f = \sum_{|i|=d} c_i X^i + \text{terms of lower degree}, \quad n > 0.$$

Here  $i = (i_1, \dots, i_n) \in \mathbb{N}^n$ ,  $|i| = i_1 + \dots + i_n$  and  $X^i = X_1^{i_1} \dots X_n^{i_n}$ ,  $c_i \in K$ . Let  $(\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$ . We are going to make the (invertible) substitution

$$\begin{aligned} X_1 &\rightarrow X_1 + \lambda_1 X_n, \\ &\vdots \\ X_{n-1} &\rightarrow X_{n-1} + \lambda_{n-1} X_n, \\ X_n &\rightarrow X_n. \end{aligned}$$

Note that, with  $\lambda^i = \lambda_1^{i_1} \dots \lambda_{n-1}^{i_{n-1}}$ , this substitution transforms  $f$  into

$$\begin{aligned} &f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n) \\ &= \left( \sum_{|i|=d} c_i \lambda^i \right) X_n^d + \text{terms of degree } < d \text{ in } X_n. \end{aligned}$$

Since the polynomial  $\sum_{|i|=d} c_i Y_1^{i_1} \dots Y_{n-1}^{i_{n-1}} \in K[Y_1, \dots, Y_{n-1}]$  is nonzero, it follows that for “almost all”  $\lambda = (\lambda_1, \dots, \lambda_{n-1}) \in K^{n-1}$ , namely for all those  $\lambda$  that are not zeros of this polynomial, the transformed polynomial

$$f(X_1 + \lambda_1 X_n, \dots, X_{n-1} + \lambda_{n-1} X_n, X_n)$$

is of (total) degree  $d$ , and has a term of the form  $c X_n^d$  with  $c \in K - \{0\}$ .